

Week 1 Assignment Report

Q1 What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, decision-making, language understanding, and problem solving.

Q2 Difference between AI, Machine Learning, Generative AI, and Agentic AI

AI is the broad field of creating intelligent systems. Machine Learning is a subset of AI that learns from data. Generative AI creates new content such as text, images, and code. Agentic AI can plan, make decisions, and perform actions autonomously to achieve goals.

Q3 Three Open-Source AI Models on Hugging Face

1. Llama 3: Large language model for text generation. 2. Mistral 7B: Efficient open-weight language model. 3. Stable Diffusion: Image generation model.

Q4 Traditional Programming vs AI-Assisted Development

Traditional programming follows explicit rules written by developers. AI-assisted development uses AI tools to generate code, suggest fixes, and automate repetitive tasks, improving productivity.

Q5 AI Hallucinations

AI hallucinations occur when an AI model generates incorrect or fabricated information. Example: An AI confidently providing a non-existent research paper citation.

AI Generated Image Placeholder

Insert any AI-generated image or screenshot here before final submission.